INTRODUCTION

Prior lessons covered pricing, yields, and spreads for bonds with fixed coupon rates and times-to-maturity of one year or longer. The next two lessons broaden the discussion to include instruments with variable rather than fixed coupons, known as floating-rate instruments, and those with original maturities of one year or less, known as money market instruments. Both types of instrument are important for investors and issuers. Floating-rate instruments, by adjusting cash flows to changes in interest rates, bear less price risk than fixed-rate instruments and are used to hedge certain exposures and to match asset and liability cash flows. Most loans are floating-rate instruments. Money market instruments are a significant source of short-term financing for many types of issuers. A short time-to-maturity means that investors can reinvest and issuers can refinance relatively quickly, which reduces interest rate risk.

Most of the examples and exhibits used throughout the reading can be downloaded as a Microsoft Excel workbook. Each worksheet in the workbook is labeled with the corresponding example or exhibit number in the text.

Learning Module Overview

- A floating-rate instrument is a debt instrument with interest determined by an observed market reference rate (MRR) plus a quoted margin. Interest payments are reset, capturing any change in the MRR, on predetermined dates.
- The quoted margin is a specified spread over or under the reference rate. The required margin, also known as the discount margin, is the spread required by investors.
- The required margin reflects “bottom-up” or issuer- and security-specific risks and is analogous to a yield spread for a fixed-rate bond discussed in prior lessons. If a floater trades at par, the quoted and required margins are equal.
- Money market instruments have original maturities of one year or less and are quoted using different conventions from those of longer-dated securities. Quotes are made on a discount rate or add-on rate basis.
- Money market discount rates are interest income divided by the face value (maturity value). They understate the investor’s rate of return if the purchase price is below the face value and vice versa.
- Conventional money market measures can be converted to enhance comparability to longer-term securities.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of the learning module.

1. The following information relates to three floating-rate notes (FRNs) issued at par value that have the three-month MRR as their reference rate.

FRN 1	140 bps	128 bps
FRN 2	145 bps	145 bps
FRN 3	150 bps	165 bps

Based on the information provided, identify which FRN will be priced at a discount on the next reset date.

A. FRN 1
B. FRN 2
C. FRN 3

Hide Solution

Solution:

C is correct. FRN 3 will be priced at a discount on the next reset date since its quoted margin of 150 bps is less than its discount margin of 165 bps. The discount amount is the present value of the deficient future interest payments of 15 bps per quarter (150 bps – 165 bps).

B is incorrect because FRN 2 will be priced at par value on the next reset date since its quoted margin is equal to its discount margin.

A is incorrect because FRN 1 will be priced at a premium because its quoted margin of 140 bps is greater than its discount margin of 128 bps. The premium amount is the present value of the extra, or "excess," interest payments of 12 bps each quarter (140 bps – 128 bps).

2. On a reset date, if the required margin falls to 30 bps because of a change in the issuer's credit risk, an FRN that has a quoted margin of 50 bps will be priced at a:

A. discount.
B. premium.
C. discount or premium, depending on its duration.

Hide Solution

Solution:

B is correct. The required margin reflects the credit risk of the issuer. It declined, meaning that the credit risk of the issuer improved; in other words, the market demands a lower spread. The required margin is below the quoted margin, so this floater will be priced at a premium.

3. A 365-day year bank certificate of deposit has an initial principal amount of USD96.5 million and a redemption amount due at maturity of USD100 million. The number of days between settlement and maturity is 270. The add-on rate of the certificate of deposit is closest to:

A. 3.63%.
B. 4.82%.
C. 4.90%.

Hide Solution

Solution:

C is correct. The add-on rate is closest to 4.90%. The add-on rate is calculated as

$$AOR = \frac{Year}{Days} \times \frac{FV - PV}{PV}.$$

$$AOR = \frac{365}{270} \times \frac{100 - 96.5}{96.5}.$$

$$AOR = 0.04903 = 4.90\%.$$

face value, its bond equivalent yield rate is closest to:

A. 0.107%.
B. 0.120%.
C. 0.135%.

Hide Solution

Solution:

B is correct. The bond equivalent yield is closest to 0.120%. First, we need to calculate the price of the instrument. Then, we can calculate the AOR or its bond equivalent yield.

The price (*PV*) of the commercial paper instrument is

$$PV = FV \times (1 - \frac{Days}{Year} \times DR).$$

$$PV = 100 \times (1 - \frac{90}{360} \times 0.0012).$$

$$PV = 99.97.$$

PV is used to calculate the add-on rate or bond equivalent yield:

$$AOR = \frac{Year}{Days} \times \frac{FV - PV}{PV}.$$

$$AOR = \frac{360}{90} \times \frac{100 - 99.97}{99.97}.$$

$$AOR = 0.0012 = 0.12\%.$$

5. A fixed-income analyst is evaluating two instruments:

- A 180-day Thai bank certificate of deposit (CD) quoted at a discount rate of 5.95%, assuming a 360-day year and a bond equivalent yield of 6.218%
- A 180-day Thai corporate commercial paper (CP) quoted as an add-on rate of 6.10%, assuming a 365-day year and a bond equivalent yield of 6.100%.

Assuming the six-month Thai T-bill is quoted at 4.36%, which instrument *most likely* has higher credit risk?

Hide Solution

Solution:

The Thai CD. The spreads over the Thai T-bill for the CD and CP are 186 bps and 174 bps, respectively. We use the bond equivalent yield measure to compare the return of each instrument and derive the spread of each one over the Thai T-bill.

The Thai bank CD has a spread of

6.218% – 4.36% = 186 bps.

And the Thai corporate CP has a spread of

6.10% – 4.36% = 174 bps.

Therefore, the Thai bank CD has a higher credit risk than the Thai corporate CP.

Discuss Filter

AK in question 5, should one not convert the yields to 360 or 365 days equivalent for comparison purposes?

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